

Analyses: Classification with KNN

01/03/2024

Description

k-nearest neighbor analyses of the feature vectors per character sequence (loaded from ~/results/features_combined.csv). The results are stored in ~/results/classifier/.

Load Packages

If the libraries are not installed yet, you need to install them using, for example, the command: `install.packages("ggplot2")`.

```
library(dplyr)
library(class)
library(gmodels)
library(caret)
```

Load Data

Load data table with estimated feature values per sequence.

```
# load feature value estimations
estimations.df <- read.csv("~/Github/UpperPalCombinatorics/results/features/features_combined.csv")
head(estimations.df)
```

```
##      subcorpus      sequence num.char huni.chars hrate.chars  ttr.chars
## 1 Aurignacien ||| ///// _ V      12  1.8962406   1.5295563  0.41666667
## 2 Aurignacien      vvvvv         5  0.0000000   0.5711903  0.20000000
## 3 Aurignacien      vvvv vvvv      9  0.5032583   0.9060047  0.22222222
## 4 Aurignacien vvvvvvvvvvvvvvv      14  0.0000000   0.6379550  0.07142857
## 5 Aurignacien      |||           3  0.0000000   0.4308271  0.33333333
## 6 Aurignacien      |||           4  0.0000000   0.5070802  0.25000000
##      rm.chars
## 1 0.6363636
## 2 1.0000000
## 3 0.7500000
## 4 1.0000000
## 5 1.0000000
## 6 1.0000000
```

Select subcorpora for binary classification.

```
# choose subcorpora to compare
unique(estimations.df$subcorpus)
```

```
## [1] "Aurignacien"      "Cuneiform (Uruk V)"  "Cuneiform (Uruk IV)"
## [4] "Cuneiform (Uruk III)" "TeDDi"
```

```
subcorpus1 <- "Cuneiform (Uruk V)"
subcorpus2 <- "Cuneiform (Uruk III)"
selected <- c(subcorpus1, subcorpus2)
estimations.subset <- estimations.df[estimations.df$subcorpus %in% selected, ]
```

Remove NAs (whole row)

```
estimations.subset <- na.omit(estimations.subset)
```

Center and scale the data

```
# first column is subcorpus column, the others are the features
# be aware that this needs to be changed in case of other table format
estimations.scaled <- cbind(estimations.subset[, c("subcorpus", "sequence", "num.char")],
                           scale(estimations.subset[, c("huni.chars", "hrate.chars",
                                                         "ttr.chars", "rm.chars")]))
```

Create Training and Test Sets

```
# Generating seed
set.seed(1234)
# Randomly generating our training and test samples with a respective ratio of 2/3 and 1/3
datasample <- sample(2, nrow(estimations.scaled), replace = TRUE, prob = c(0.67, 0.33))

# Generate training set (select rows randomly by using datasample vector)
train <- estimations.scaled[datasample == 1, 1:ncol(estimations.scaled)]
# Generate test set
test <- estimations.scaled[datasample == 2, 1:ncol(estimations.scaled)]
```

Initialize data frame

```
knn.results <- data.frame(k = numeric(0), accuracy = numeric(0), precision = numeric(0),
                          recall = numeric(0), f1 = numeric(0))
```

Building knn classifier

```
# choose maximum number of neighbors n
n = 10
# run a loop over different numbers of neighbors up to n
for (k in 1:n){
  # knn estimation of labels
  predictions.knn <- knn(train = train[, c("huni.chars",
                                           "hrate.chars",
                                           "ttr.chars",
                                           "rm.chars")],
```

```

        test = test[, c("huni.chars",
                        "hrate.chars",
                        "ttr.chars",
                        "rm.chars")],
        cl = train[, "subcorpus"], k = k)

# model evaluation
# creating a dataframe from known (true) test labels
test.labels <- test[, "subcorpus"]
# combining predicted and known classes
class.comparison <- data.frame(predictions.knn, test.labels)
# giving appropriate column names
names(class.comparison) <- c("predicted", "observed")
# inspecting our results table
head(class.comparison)
# get confusion matrix
cm <- confusionMatrix(data = class.comparison$predicted,
                      reference = as.factor(class.comparison$observed))

print(cm)
# get precision, recall, and f1 from the output list of confusionMatrix()
accuracy <- cm$overall['Accuracy']
f1 <- cm[["byClass"]][["F1"]]
recall <- cm[["byClass"]][["Recall"]]
precision <- cm[["byClass"]][["Precision"]]

# prepare data frame with results
local.results <- data.frame(k, accuracy, precision, recall, f1, row.names = NULL)
local.results.rounded <- round(local.results, 2)
# print(local.results.rounded)
knn.results <- rbind(knn.results, local.results.rounded)
# give predictions alongside the other information in data frame
knn.predictions.df <- cbind(test, predictions.knn)
}

```

```

## Confusion Matrix and Statistics
##
##               Reference
## Prediction      Cuneiform (Uruk III) Cuneiform (Uruk V)
## Cuneiform (Uruk III)                274                8
## Cuneiform (Uruk V)                  13                17
##
##               Accuracy : 0.9327
##               95% CI : (0.899, 0.9579)
##      No Information Rate : 0.9199
##      P-Value [Acc > NIR] : 0.2369
##
##               Kappa : 0.5816
##
## Mcnemar's Test P-Value : 0.3827
##
##               Sensitivity : 0.9547
##               Specificity : 0.6800
##      Pos Pred Value : 0.9716
##      Neg Pred Value : 0.5667

```

```

##           Prevalence : 0.9199
##           Detection Rate : 0.8782
##           Detection Prevalence : 0.9038
##           Balanced Accuracy : 0.8174
##
##           'Positive' Class : Cuneiform (Uruk III)
##
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##           'Positive' Class : Cuneiform (Uruk III)
##
## Confusion Matrix and Statistics
##
##               Reference
## Prediction      Cuneiform (Uruk III) Cuneiform (Uruk V)
## Cuneiform (Uruk III)                274                6
## Cuneiform (Uruk V)                  13               19
##
##           Accuracy : 0.9391
##           95% CI : (0.9065, 0.9629)
##           No Information Rate : 0.9199
##           P-Value [Acc > NIR] : 0.1233
##
##           Kappa : 0.6337
##
## Mcnemar's Test P-Value : 0.1687
##
##           Sensitivity : 0.9547
##           Specificity : 0.7600
##           Pos Pred Value : 0.9786
##           Neg Pred Value : 0.5938

```

```

##           Prevalence : 0.9199
##           Detection Rate : 0.8782
##           Detection Prevalence : 0.8974
##           Balanced Accuracy : 0.8574
##
##           'Positive' Class : Cuneiform (Uruk III)
##
## Confusion Matrix and Statistics
##
##               Reference
## Prediction      Cuneiform (Uruk III) Cuneiform (Uruk V)
## Cuneiform (Uruk III)                275                5
## Cuneiform (Uruk V)                  12               20
##
##           Accuracy : 0.9455
##           95% CI : (0.9142, 0.9679)
##           No Information Rate : 0.9199
##           P-Value [Acc > NIR] : 0.05316
##
##           Kappa : 0.6723
##
## Mcnemar's Test P-Value : 0.14561
##
##           Sensitivity : 0.9582
##           Specificity : 0.8000
##           Pos Pred Value : 0.9821
##           Neg Pred Value : 0.6250
##           Prevalence : 0.9199
##           Detection Rate : 0.8814
##           Detection Prevalence : 0.8974
##           Balanced Accuracy : 0.8791
##
##           'Positive' Class : Cuneiform (Uruk III)
##
## Confusion Matrix and Statistics
##
##               Reference
## Prediction      Cuneiform (Uruk III) Cuneiform (Uruk V)
## Cuneiform (Uruk III)                274                5
## Cuneiform (Uruk V)                  13               20
##
##           Accuracy : 0.9423
##           95% CI : (0.9104, 0.9655)
##           No Information Rate : 0.9199
##           P-Value [Acc > NIR] : 0.08303
##
##           Kappa : 0.6585
##
## Mcnemar's Test P-Value : 0.09896
##
##           Sensitivity : 0.9547
##           Specificity : 0.8000
##           Pos Pred Value : 0.9821
##           Neg Pred Value : 0.6061

```

```

##           Prevalence : 0.9199
##           Detection Rate : 0.8782
##           Detection Prevalence : 0.8942
##           Balanced Accuracy : 0.8774
##
##           'Positive' Class : Cuneiform (Uruk III)
##
## Confusion Matrix and Statistics
##
##               Reference
## Prediction      Cuneiform (Uruk III) Cuneiform (Uruk V)
## Cuneiform (Uruk III)                274                5
## Cuneiform (Uruk V)                  13               20
##
##           Accuracy : 0.9423
##           95% CI : (0.9104, 0.9655)
##           No Information Rate : 0.9199
##           P-Value [Acc > NIR] : 0.08303
##
##           Kappa : 0.6585
##
## Mcnemar's Test P-Value : 0.09896
##
##           Sensitivity : 0.9547
##           Specificity : 0.8000
##           Pos Pred Value : 0.9821
##           Neg Pred Value : 0.6061
##           Prevalence : 0.9199
##           Detection Rate : 0.8782
##           Detection Prevalence : 0.8942
##           Balanced Accuracy : 0.8774
##
##           'Positive' Class : Cuneiform (Uruk III)
##
## Confusion Matrix and Statistics
##
##               Reference
## Prediction      Cuneiform (Uruk III) Cuneiform (Uruk V)
## Cuneiform (Uruk III)                274                5
## Cuneiform (Uruk V)                  13               20
##
##           Accuracy : 0.9423
##           95% CI : (0.9104, 0.9655)
##           No Information Rate : 0.9199
##           P-Value [Acc > NIR] : 0.08303
##
##           Kappa : 0.6585
##
## Mcnemar's Test P-Value : 0.09896
##
##           Sensitivity : 0.9547
##           Specificity : 0.8000
##           Pos Pred Value : 0.9821
##           Neg Pred Value : 0.6061

```

```

##           Prevalence : 0.9199
##           Detection Rate : 0.8782
##           Detection Prevalence : 0.8942
##           Balanced Accuracy : 0.8774
##
##           'Positive' Class : Cuneiform (Uruk III)
##
## Confusion Matrix and Statistics
##
##               Reference
## Prediction      Cuneiform (Uruk III) Cuneiform (Uruk V)
## Cuneiform (Uruk III)                274                3
## Cuneiform (Uruk V)                  13               22
##
##           Accuracy : 0.9487
##           95% CI : (0.9181, 0.9704)
##           No Information Rate : 0.9199
##           P-Value [Acc > NIR] : 0.03223
##
##           Kappa : 0.7058
##
## Mcnemar's Test P-Value : 0.02445
##
##           Sensitivity : 0.9547
##           Specificity : 0.8800
##           Pos Pred Value : 0.9892
##           Neg Pred Value : 0.6286
##           Prevalence : 0.9199
##           Detection Rate : 0.8782
##           Detection Prevalence : 0.8878
##           Balanced Accuracy : 0.9174
##
##           'Positive' Class : Cuneiform (Uruk III)
##
## Confusion Matrix and Statistics
##
##               Reference
## Prediction      Cuneiform (Uruk III) Cuneiform (Uruk V)
## Cuneiform (Uruk III)                275                4
## Cuneiform (Uruk V)                  12               21
##
##           Accuracy : 0.9487
##           95% CI : (0.9181, 0.9704)
##           No Information Rate : 0.9199
##           P-Value [Acc > NIR] : 0.03223
##
##           Kappa : 0.6965
##
## Mcnemar's Test P-Value : 0.08012
##
##           Sensitivity : 0.9582
##           Specificity : 0.8400
##           Pos Pred Value : 0.9857
##           Neg Pred Value : 0.6364

```

```

##           Prevalence : 0.9199
##           Detection Rate : 0.8814
##           Detection Prevalence : 0.8942
##           Balanced Accuracy : 0.8991
##
##           'Positive' Class : Cuneiform (Uruk III)
##
## Confusion Matrix and Statistics
##
##               Reference
## Prediction      Cuneiform (Uruk III) Cuneiform (Uruk V)
## Cuneiform (Uruk III)                275                 3
## Cuneiform (Uruk V)                  12                22
##
##           Accuracy : 0.9519
##           95% CI : (0.9219, 0.9728)
##           No Information Rate : 0.9199
##           P-Value [Acc > NIR] : 0.01843
##
##           Kappa : 0.7199
##
## Mcnemar's Test P-Value : 0.03887
##
##           Sensitivity : 0.9582
##           Specificity : 0.8800
##           Pos Pred Value : 0.9892
##           Neg Pred Value : 0.6471
##           Prevalence : 0.9199
##           Detection Rate : 0.8814
##           Detection Prevalence : 0.8910
##           Balanced Accuracy : 0.9191
##
##           'Positive' Class : Cuneiform (Uruk III)
##

```

```
print(knn.results)
```

```

##      k accuracy precision recall  f1
## 1  1    0.93    0.97    0.95 0.96
## 2  2    0.93    0.97    0.95 0.96
## 3  3    0.94    0.98    0.95 0.97
## 4  4    0.95    0.98    0.96 0.97
## 5  5    0.94    0.98    0.95 0.97
## 6  6    0.94    0.98    0.95 0.97
## 7  7    0.94    0.98    0.95 0.97
## 8  8    0.95    0.99    0.95 0.97
## 9  9    0.95    0.99    0.96 0.97
## 10 10    0.95    0.99    0.96 0.97

```

Write to file.

```

# write results
write.csv(knn.results, file = paste(c("~/Github/UpperPalCombinatorics/results/classifier/KNN/results_KNN",
                                     subcorpus1, "_",
                                     subcorpus2, "_",
                                     ".csv")

```

```

                                ), collapse = ""),
                                row.names = F)

# write df with predictions for post-hoc analyses
# note that this writes the predictions only for the last run of the loop, i.e. the highest k
write.csv(knn.predictions.df, file = paste(c("~/Github/UpperPalCombinatorics/results/classifier/KNN/pre
                                subcorpus1, "_",
                                subcorpus2, "_",
                                ".csv"
                                ), collapse = ""),
                                row.names = F)

```